

LUGANO, Switzerland

WMO: 067700

Lat: 46.0039N

Lon: 8.9601E

Elev: 279

StdP: 98.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.3	(c) -1.1	(d) -16.9	(e) 0.9	(f) 4.7	(g) -13.9	(h) 1.2	(i) 6.6	(j) 9.4	(k) 8.4	(l) 8.4	(m) 7.7	(n) 1.4	(o) 340	(p) 0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.3	(c) 30.3	(d) 21.7	(e) 29.0	(f) 21.0	(g) 27.8	(h) 20.4	(i) 22.9	(j) 28.9	(k) 22.1	(l) 27.6	(m) 21.3	(n) 26.6	(o) 2.4	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	16.1	26.6	20.1	15.3	25.6	19.3	14.6	24.8	69.2	29.1	65.9	27.7	63.2	26.7	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.0	(b) 6.5	(c) 5.2	(e) -4.9	(f) 32.5	(g) 2.1	(h) 1.3	(i) -6.5	(j) 33.5	(k) -7.7	(l) 34.2	(m) -8.9	(n) 35.0	(o) -10.4	(p) 36.0		
(4) 8.0	(5) 6.5	(5) 5.2	(4) -4.9	(4) 32.5	(4) 2.1	(4) 1.3	(4) -6.5	(4) 33.5	(4) -7.7	(4) 34.2	(4) -8.9	(4) 35.0	(4) -10.4	(4) 36.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	3.9	5.2	9.1	12.4	16.6	20.7	22.7	22.1	18.0	13.7	8.5	4.7	(6)
(7)		DBStd	7.17	2.38	2.67	3.14	3.07	2.93	3.04	2.49	2.43	2.75	2.79	2.86	2.57	(7)
(8)		HDD10.0	621	190	136	53	11	1	0	0	0	0	4	61	166	(8)
(9)		HDD18.3	2264	448	368	285	178	68	11	2	2	37	145	296	424	(9)
(10)		CDD10.0	1783	1	1	26	84	206	322	394	376	241	119	14	1	(10)
(11)		CDD18.3	386	0	0	0	1	16	83	137	120	28	1	0	0	(11)
(12)		CDH23.3	2417	0	0	3	6	77	529	966	729	102	4	0	0	(12)
(13)		CDH26.7	541	0	0	0	2	7	118	239	167	7	1	0	0	(13)
(14)	Wind	WSAvg	1.6	1.4	1.6	2.0	1.9	1.7	1.7	1.6	1.5	1.4	1.4	1.4	(14)	
(15)	Precipitation	PrecAvg	1589	76	70	94	139	193	166	152	163	169	149	146	70	(15)
(16)		PrecMax	2787	279	277	316	512	426	444	397	489	447	540	587	198	(16)
(17)		PrecMin	979	0	0	2	9	49	17	14	12	10	0	0	1	(17)
(18)		PrecStd	368	66	64	72	89	90	89	95	94	122	118	120	53	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	16.9	22.3	23.7	27.5	31.6	31.8	32.2	27.6	23.2	16.8	14.2	(19)
(20)			MCWB	7.3	7.7	10.7	14.2	18.2	21.6	22.2	22.7	20.3	16.6	10.6	5.7	(20)
(21)		2%	DB	11.0	13.1	18.7	21.4	25.4	29.6	30.3	30.1	26.0	20.8	14.9	11.2	(21)
(22)			MCWB	4.5	6.1	10.0	13.1	17.1	21.1	21.7	21.9	19.2	15.8	10.7	6.2	(22)
(23)		5%	DB	9.0	11.0	16.4	19.7	23.7	27.9	29.1	28.4	24.3	19.2	13.6	9.6	(23)
(24)			MCWB	4.4	5.6	9.1	12.4	16.2	20.4	21.2	20.9	18.2	14.9	10.2	6.4	(24)
(25)		10%	DB	7.5	9.4	14.6	18.0	22.0	26.4	28.0	27.1	22.9	17.9	12.5	8.5	(25)
(26)			MCWB	4.4	4.9	8.4	11.5	15.4	19.5	20.5	20.3	17.4	14.6	10.0	5.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	9.1	12.2	15.7	19.7	23.7	24.2	23.9	21.5	18.1	13.4	9.6	(27)
(28)			MCDWB	12.6	14.0	19.2	21.8	25.4	29.5	29.7	30.3	26.3	21.1	14.6	10.7	(28)
(29)		2%	WB	6.8	7.7	11.0	14.4	18.4	22.2	23.1	22.7	20.3	17.0	12.3	8.2	(29)
(30)			MCDWB	8.8	11.0	16.3	19.6	23.8	27.8	29.0	28.7	24.7	19.5	13.6	9.3	(30)
(31)		5%	WB	5.9	6.7	10.2	13.4	17.3	21.3	22.2	21.9	19.4	16.1	11.6	7.2	(31)
(32)			MCDWB	7.6	9.2	14.5	17.9	22.2	26.7	27.7	27.4	23.1	18.3	12.8	8.6	(32)
(33)		10%	WB	5.0	5.9	9.4	12.5	16.4	20.4	21.4	21.1	18.4	15.2	10.7	6.4	(33)
(34)			MCDWB	6.8	8.1	13.2	16.4	20.6	25.4	26.8	26.1	21.8	17.3	11.9	8.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	7.4	8.7	8.3	8.5	9.0	9.3	9.0	8.3	7.0	6.2	6.0	(35)
(36)			MCDBR	9.4	10.8	12.8	11.6	11.2	10.6	10.6	10.8	10.1	9.0	7.8	7.6	(36)
(37)			MCWBR	5.4	6.1	6.3	5.4	5.5	5.3	5.3	5.3	5.3	5.3	5.0	4.7	(37)
(38)		5% WB	MCDBR	7.2	9.1	11.0	9.3	9.6	9.6	9.6	9.7	8.8	7.2	5.5	5.6	(38)
(39)			MCWBR	4.7	5.6	5.8	4.6	5.0	5.1	5.3	5.1	4.9	4.5	3.9	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.322	0.359	0.417	0.477	0.484	0.502	0.484	0.475	0.458	0.428	0.364	0.319	(40)	
(41)		taud	2.308	2.233	2.130	2.014	2.047	2.052	2.115	2.149	2.153	2.188	2.318	2.361	(41)	
(42)		Ebn at Noon	761	798	797	782	791	777	786	776	749	707	702	725	(42)	
(43)		Edh at Noon	79	104	133	163	163	163	151	141	129	108	79	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.53	2.41	3.69	4.49	5.16	6.01	6.07	5.20	3.97	2.48	1.50	1.27	(44)	
(45)		RadStd	0.20	0.31	0.50	0.62	0.47	0.44	0.45	0.39	0.33	0.34	0.24	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.48	+0.87	N/A	N/A	N/A	N/A	-82	-157	+154	+84
(47) Regional (109 neighbors)		N/A	N/A	N/A	+0.72	+0.41	+0.32	N/A	N/A	+59	+16

Nomenclature: See separate page